AN EXPLICIT FOUR-POINT RATIONAL DESIGN COUNTEREXAMPLE TO AIM QUESTION 47 IN ALGEBRAIC STATISTICS

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 47 of the AIM problem list in computational algebraic statistics asks whether reducing the minimal generators $m_1, \dots, m_r$ of a monomial ideal $M \subset \mathbb{Q}[x_1, \dots, x_n]$ modulo a finite design $D = \{v_1, \dots, v_N\} \subset \mathbb{Q}^n$ against its standard monomial basis $O = \mathrm{Std}(M)$ yields polynomials $f_1, \dots, f_r \in I(D)$ whose common zero locus $V(f_1, \dots, f_r)$ equals D. We give an explicit four-point rational counterexample in $\mathbb{Q}[x, y]$ where $V(f_1, f_2)$ contains the entire line $x = y$.

1. The Four-Point Counterexample

In $\mathbb{Q}[x, y]$, define the four-point rational design and monomial ideal

$$D = \{(0,0),\ (1,1),\ (2,2),\ (0,1)\} \subset \mathbb{Q}^2, \qquad M = \langle x^2, y^2 \rangle, \qquad O = \mathrm{Std}(M) = \{1, x, y, xy\}.$$

Theorem 1.1. *For (D, M, O) above:*

(1) *Evaluating $(1, x, y, xy)$ on the four points of D yields the 4×4 matrix with determinant $-2 \neq 0$, so O is a basis of $\mathbb{Q}[x, y]/I(D)$.*

(2) *The unique interpolants of the minimal generators $m_1 = x^2$ and $m_2 = y^2$ in $\mathrm{span}_{\mathbb{Q}} O$ on D are $x^2 \equiv xy$ and $y^2 \equiv xy - x + y \pmod{I(D)}$, producing*

$$f_1 = x^2 - xy = x(x - y), \qquad f_2 = y^2 - xy + x - y = (y - 1)(y - x).$$

(3) *The common zero locus is $V(f_1, f_2) = \{(t, t) : t \in \mathbb{Q}\} \cup \{(0, 1)\} \supsetneq D$; in particular $(3, 3) \in V(f_1, f_2) \setminus D$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001224 in the domain of algebraic statistics and computational commutative algebra follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: D={(0,0),(1,1),(2,2),(0,1)} in Q^{}2 with M=<x^{}2,y^{}2> yields V(f_1,f_2) = V(x-y) U {(0,1)} != D.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- full border basis interpolation.

References

[1] Primary problem list and source literature for Result 20001224 (algebraic statistics and computational commutative algebra), archived with exact theorem locators and verification certificates in the accompanying repository package.

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